

Q1

YAMANURI PRANEETH
192372269RA

SIMATS Online Programming Lab

JAVA PROGRAMMING

Your Code

```
1 import java.util.*;
2
3 public class Main {
4
5     static class Patient {
6         String patientId;
7         String name;
8         String mobile;
9         String department;
10        String doctor;
11
12        Patient(String patientId, String name, String mobile, String d
13            this.patientId = patientId;
14            this.name = name;
15            this.mobile = mobile;
16            this.department = department;
17            this.doctor = doctor;
18        }
19    }
20
21    public static void main(String[] args) {
22        Scanner sc = new Scanner(System.in);
23
24        HashMap<String, Patient> patientMap = new HashMap<>();
25        ArrayList<Patient> patientList = new ArrayList<>();
26        HashMap<String, Integer> deptCount = new HashMap<>();
27
28        int maxPatients = 100;
29
30        while (true) {
31            String line = nextNonEmptyLine(sc);
32            if (line == null) break; // no more input
33            int choice;
34            try {
35                choice = Integer.parseInt(line.trim());
```

Input

```
1
P101
Rahul Kumar
9876543210
```

Output

```
Name : Rahul Kumar
Mobile : 9876543210
Department : CARDIOLOGY
Doctor : Dr.Arun
```

Visible Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

0.00 / 20 marks

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```
46        String patientNameRaw = sc.nextLine();
47        String mobile = sc.nextLine();
48        String departmentRaw = sc.nextLine();
49        String doctorRaw = sc.nextLine();
50
51        patientNameRaw = removeExtraSpaces(patientNameRaw);
52        mobile = removeExtraSpaces(mobile).replaceAll("\\s+", "");
53        departmentRaw = removeExtraSpaces(departmentRaw);
54        doctorRaw = removeExtraSpaces(doctorRaw);
55
56        if (patientMap.containsKey(patientId)) {
57            continue;
58        }
59
60        String nameNoSpaces = patientNameRaw.replaceAll("\\s+", "");
61        if (!nameNoSpaces.matches("[A-Za-z]+")) continue;
62        if (!mobile.matches("\\d{10}")) continue;
63
64        String department = departmentRaw.toUpperCase();
65        String properName = toProperCase(patientNameRaw);
66
67        Patient p = new Patient(patientId, properName, mobile,
68            patientMap.put(patientId, p);
69            patientList.add(p);
70
71        deptCount.put(department, deptCount.getOrDefault(depart
72
73        System.out.println("Patient Registered Successfully.");
74
75    } else if (choice == 2) {
76        String id = sc.nextLine().trim();
77        Patient p = patientMap.get(id);
78
79        if (p == null) {
80            System.out.println("Patient Not Found");
81        } else {
82            System.out.println("Patient Details");
```

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```
1110     sc.close();
1111 }
1112
1113 private static String nextNonEmptyLine(Scanner sc) {
1114     while (sc.hasNextLine()) {
1115         String s = sc.nextLine();
1116         if (s == null) return null;
1117         s = s.trim();
1118         if (!s.isEmpty()) return s;
1119     }
1120     return null;
1121 }
1122
1123 private static String removeExtraSpaces(String s) {
1124     if (s == null) return "";
1125     return s.trim().replaceAll("\\s+", " ");
1126 }
1127
1128 private static String toProperCase(String s) {
1129     s = removeExtraSpaces(s);
1130     if (s.isEmpty()) return s;
1131
1132     String[] parts = s.split(" ");
1133     StringBuilder sb = new StringBuilder();
1134
1135     for (String w : parts) {
1136         if (w.isEmpty()) continue;
1137         String lower = w.toLowerCase();
1138         String proper = Character.toUpperCase(lower.charAt(0)) + lower.substring(1);
1139         if (sb.length() > 0) sb.append(" ");
1140         sb.append(proper);
1141     }
1142     return sb.toString();
1143 }
1144 }
1145 }
```

Q2

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JAVA PROGRAMMING

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192372269RA

Your Code

```
1 import java.util.*;
2
3 public class Main {
4
5     static class Book {
6         String bookId;
7         String title;
8         String author;
9         String category;
10        String publisher;
11
12        Book(String bookId, String title, String author, String category, String publisher) {
13            this.bookId = bookId;
14            this.title = title;
15            this.author = author;
16            this.category = category;
17            this.publisher = publisher;
18        }
19    }
20
21    static String toTitleCase(String s) {
22        s = s.trim().toLowerCase();
23        if (s.isEmpty()) return s;
24
25        StringBuilder sb = new StringBuilder();
26        String[] parts = s.split("\\s+");
27        for (int i = 0; i < parts.length; i++) {
28            String w = parts[i];
29            if (w.length() == 0) continue;
30            sb.append(Character.toUpperCase(w.charAt(0))).append(w.substring(1));
31            if (i < parts.length - 1) sb.append(" ");
32        }
33        return sb.toString();
34    }
35}
```

Input

```
1
8101
java programming
Herbert Schildt
```

Output

```
----- Online Library Book
Management System -----
1. Add Book
2. Search Book by ID
3. Display All Books
4. Display All Books
```

Visible Test Cases

Test Case 1	✖
Test Case 2	✖
Test Case 3	✖

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JAVA PROGRAMMING

```
41         return s;
42     }
43
44     public static void main(String[] args) {
45         Scanner sc = new Scanner(System.in);
46
47         ArrayList<Book> books = new ArrayList<>(500);
48         HashMap<String, Book> bookById = new HashMap<>();
49         HashSet<String> uniqueAuthors = new HashSet<>();
50         HashMap<String, Integer> categoryCount = new HashMap<>();
51
52         while (true) {
53             System.out.println("----- Online Library Book Management System -----");
54             System.out.println("1. Add Book");
55             System.out.println("2. Search Book by ID");
56             System.out.println("3. Display All Books");
57             System.out.println("4. Display Unique Authors");
58             System.out.println("5. Category-wise Book Count");
59             System.out.println("6. Exit");
60             System.out.print("Enter choice: ");
61
62             int choice;
63             try {
64                 choice = Integer.parseInt(sc.nextLine().trim());
65             } catch (Exception e) {
66                 System.out.println("Invalid choice. Please enter a number between 1 and 6.");
67                 continue;
68             }
69
70             if (choice == 1) {
71                 if (books.size() >= 500) {
72                     System.out.println("Maximum book limit reached.");
73                     continue;
74                 }
75
76                 System.out.print("Enter Book ID: ");
77                 String bookId = sc.nextLine().trim();
```

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```
123             System.out.println("Category : " + b.category);
124         } else {
125             System.out.println("Book Not Available");
126         }
127         System.out.println();
128
129         } else if (choice == 3) {
130             System.out.println("-----");
131             System.out.println("ID\tTITLE\t\tAUTHOR\tCATEGORY");
132             System.out.println("-----");
133
134             for (Book b : books) {
135                 System.out.println(b.bookId + "\t" + b.title + "\t" + b.author + "\t" + b.category);
136             }
137             System.out.println();
138
139         } else if (choice == 4) {
140             System.out.println("Unique Authors");
141             for (String a : uniqueAuthors) {
142                 System.out.println(a);
143             }
144             System.out.println();
145
146         } else if (choice == 5) {
147             System.out.println("Category-wise Count");
148             for (Map.Entry<String, Integer> e : categoryCount.entrySet()) {
149                 System.out.println(e.getKey() + " : " + e.getValue());
150             }
151             System.out.println();
152
153         } else if (choice == 6) {
154             System.out.println("Exiting... Thank you!");
155             break;
156
157         } else {
158             System.out.println("Invalid choice. Please enter a number between 1 and 6.");
159             System.out.println();
160         }
```